RP-Cloud Labs: Tool-Session Service Functional Requirements

Functional requirements: describe what the RP-Cloud Labs Tool-Session Service must do to allow students to request hosted software tools and enable lecturers to monitor students' practical sessions..

1. Request a tool session: The system shall allow a registered student to submit a tool-session request by selecting a hosted software tool and specifying the requested session duration.

2. Validate the student account: The system shall verify that the student's account is registered, active, and validated when the student submits a tool-session request by selecting a hosted software tool and specifying the requested session duration..

3. Check license capacity: The system shall check whether the college has an available license for the hosted software tool selected by the student for the requested session duration before approving the tool-session request.

4. Record the tool session

The system shall record each approved tool-session request, including the student ID, selected hosted software tool, requested session duration, session start time, session status, and session end time.

2. Check Against Good Requirement Qualities

Each requirement is checked against five qualities: clear, complete, consistent, testable, and measurable.

Functional requirement	Clear	Complete	Consistent	Testable	Measurable
Request a tool session	Yes	Yes	Review	Yes	Review
Validate the student account	Yes	Review	Yes	Yes	Review
Check license capacity	Yes	Review	Review	Yes	Review
Record the tool session	Yes	Review	Yes	Review	Review

4. Rewritten requirements

1.**Request a tool session**: The system shall display a confirmation message after successfully receiving a tool-session request containing a valid tool name and session duration.

2**.Validate the student account**: The system shall reject a tool-session request if the student's account is not registered, active, and validated.

3.Check license capacity: The system shall approve a tool-session request only when at least one license for the selected tool is available.

4.**Record the tool session**: The system shall save a record for every approved session containing the student ID, tool name, start time, status, and end time, with the end time updated when the session ends.

4. Conclusion

The four functional requirements describe the main operations of the RP-Cloud Labs Tool-Session Service. The quality checks show that some requirements need further review, particularly their completeness, to ensure that the system's behavior is fully specified and unambiguous.